Nonlinear Dynamics and Chaos - Chapter 2: Flows on the Line

A Geometric Way of Thinking:

Consider $\dot{x} = \sin(x)$ as a flow on the x-axis in the next three problems.

Problem 2.1.1 Find all the fixed points on the flow.

Fixed points occur when $\sin(x) = 0$, so $x = n\pi$ for $n \in \mathbb{Z}$ are the fixed points of this system.

Problem 2.1.2 At which points x does the flow have the greatest velocity to the right?

Since $\sin(x) > 0$ for some points x, the flow to the right is greatest when $\sin(x)$ attains its global maximum. So $x = 2\pi n + \frac{\pi}{2}$ for $n \in \mathbb{Z}$.

Problem 2.1.3a Find the flow's acceleration $\ddot{x}$ as a function of x.

$$\frac{dx}{dt} = \sin(x) \implies \frac{d^2x}{dt^2} = \frac{d}{dt}\sin(x) = \cos(x)\frac{dx}{dt} = \cos(x)\sin(x)$$

Thus we have $\ddot{x} = \sin(x)\cos(x)$.

Problem 2.1.3b Find the points where the flow has maximum positive acceleration.

First, we find the local extrema of $\ddot{x}$ as a function of x.

$$\frac{d}{dx}\ddot{x} = [\cos(x)]\cos(x) + \sin(x)[-\sin(x)] = \cos^2(x) - \sin^2(x)$$

$$\frac{d}{dx}\ddot{x} = 0 \implies \cos^2(x) = \sin^2(x) \implies \cos(x) = \pm\sin(x)$$

The local maximums occur when x satisfies $\cos(x) = \sin(x)$, and the local minimums occur when x satisfies $\cos(x) = -\sin(x)$. We are interested in the former. Thus, $x = n\pi + \frac{\pi}{4}$, for $n \in \mathbb{Z}$. The maximum positive acceleration is $\ddot{x} = \frac{1}{2}$.

Problem 2.4.1a The exact solution of $\dot{x} = \sin(x)$, where $x(0) = x_0$, is:

$$t = \ln\left|\frac{\csc(x_0) + \cot(x_0)}{\csc(x) + \cot(x)}\right|.$$

Given the initial condition $x_0 = \frac{\pi}{4}$, show that

$$x(t) = 2\tan^{-1}\left(\frac{e^t}{1+\sqrt{2}}\right).$$

For $x_0 = \frac{\pi}{4}$, we have

$$t = \ln\left|\frac{1+\sqrt{2}}{\csc(x) + \cot(x)}\right| \implies e^t = \frac{1+\sqrt{2}}{\csc(x) + \cot(x)} \implies \frac{1}{\csc(x) + \cot(x)} = \frac{e^t}{1+\sqrt{2}}.$$

So we wish to show that

$$\frac{1}{\csc(x) + \cot(x)} = \tan(\frac{x}{2}).$$

Recall the double angle identities

$$\sin(2x) = 2\sin(x)\cos(x),$$

$\cos(2x) = 2\cos^2(x) - 1.$

Thus we have

$$\frac{1}{\csc(x) + \cot(x)} = \frac{1}{\frac{1}{\sin(x)} + \frac{\cos(x)}{\sin(x)}} = \frac{\sin(x)}{1 + \cos(x)} = \frac{2\sin(\frac{x}{2})\cos(\frac{x}{2})}{2\cos^2(\frac{x}{2})} = \frac{\sin(\frac{x}{2})}{\cos(\frac{x}{2})} = \tan(\frac{x}{2}).$$

Finally,

$$\tan(\frac{x}{2}) = \frac{e^t}{1+\sqrt{2}} \implies x(t) = 2\tan^{-1}\left(\frac{e^t}{1+\sqrt{2}}\right).$$

Problem 2.1.4b Find the analytical solution for $x(t)$ given an arbitrary initial condition x_0.

We can simply write

$$x(t) = 2\tan^{-1}\left(\frac{e^t}{\csc(x_0) + \cot(x_0)}\right).$$

Fixed Points and Stability:

In each of the following equations, find all fixed points, classify their stability, and find an analytic solution, if possible.

Problem 2.2.1 $\dot{x} = 4x^2 - 16$

$\dot{x} = 4(x-2)(x+2)$, so the fixed points are $x^* = \pm 2$. Considering the sign of $f'(x^*)$ for each, $x^* = -2$ is stable, and $x^* = 2$ is unstable. We have

$$\frac{dx}{dt} = 4x^2 - 16 \implies \frac{1}{4}\int \frac{dx}{x^2-4} = \int dt \implies t = \frac{1}{16}\left(\int \frac{dx}{x-2} - \int \frac{dx}{x+2}\right)$$

$$\implies t = \frac{1}{16}(\ln|x-2| - \ln|x+2|) + C \implies t = t = \frac{1}{16}\ln\left|\frac{x-2}{x+2}\right| + C$$

Suppose that $x(0) = x_0$. Then we have

$$C = \frac{1}{16}\ln\left|\frac{x_0+2}{x_0-2}\right|$$

So we have

$$t = \frac{1}{16}\ln\left|\frac{(x_0+2)(x-2)}{(x_0-2)(x+2)}\right|$$

Let $K = \frac{(x_0-2)}{(x_0+2)}$. Rearranging in terms of x, we get

$$\frac{(x-2)}{K(x+2)} = e^{16t} \implies x - 2 = Ke^{16t}(x+2) \implies (1 - Ke^{16t})x = 2(Ke^{16t} + 1)$$

Thus an explicit solution is

$$x(t) = 2\left(\frac{Ke^{16t} + 1}{1 - Ke^{16t}}\right)$$

Problem 2.2.7 $\dot{x} = e^x - \cos(x)$

There are an infinite number of fixed points, i.e. solutions to $e^x - \cos(x) = 0$. The rightmost solution is $x^* = 0$.

$$\frac{d}{dx}(e^x - \cos(x)) = e^x + \sin(x)$$

So at $x^* = 0$, $\dot{x} > 0$, and thus $x^* = 0$ is unstable. We can also make some qualitative statements about the infinite number of fixed points for $x^* < 0$. Since $e^x \to 0$ as $x \to -\infty$, the fixed points are better approximated by solutions to $\cos(x) = 0$ at $x \to -\infty$. Furthermore, since $\cos(x)$ oscillates continuously about the x-axis, the stability of these fixed points will alternate between stable and unstable. Since $\frac{d}{dx}e^x \to 0$ as $x \to \infty$, and $\left|\frac{d}{dx}\cos(x)\right|_{x=x^*} = 1$, the characteristic time scale $\frac{1}{|f'(x^*)|} \to 1$ as $x \to \infty$. There is no explicit analytic solution to this differential equation.

Population Growth:

Linear Stability Analysis:

Existence and Uniqueness:

Impossibility of Oscillations:

Potentials:

Solving Equations on the Computer: